CATERPILLAR® MACHINE DIAGNOSTICS

INDUSTRIAL INSPECTION & SYSTEM CONFIGURATION AUDIT

Machine ID:	CAT-GEN-C175	**Date of Inspection:**	2026-07-20
Machine Name:	CAT Diesel Generator	**Lead Engineer:**	John Administrator
Manufacturer:	Caterpillar Inc.	**Model / Category:**	C175-16 / CAT Diesel Generator
Diagnostic Status:	**Fault** (Score: 0%)	**Engine Version:**	v2.4-deterministic

System Hardware & Firmware Configuration Audit

Below is a side-by-side comparison of the active telemetry and local software modules against original factory reference parameters.

Parameter	Factory Reference	Current Machine	Audit Status
Firmware Version	v9.4.0	v9.0.2	**MISMATCH (CRITICAL)**
PLC Version	v8.32	v8.00	**MISMATCH (WARNING)**
CPU Architecture	Intel Core i3-7100U	Intel Pentium 4415U	**MISMATCH (CRITICAL)**
RAM (Memory)	16GB DDR4	8GB DDR4	**MISMATCH (WARNING)**
Storage	128GB SSD	128GB SSD	**MATCHED**
Sensor Count	24	20	**MISMATCH (WARNING)**
Communication Ports	USB, COM1, COM2, Ethernet, Modbus-TCP, Profibus	USB, COM1, Ethernet, Modbus-TCP	**MISMATCH (CRITICAL)**
Installed Modules	Analog Input, Digital IO, CAN Bus controller, Power Logger Card	Analog Input, Digital IO, CAN Bus controller	**MISMATCH (CRITICAL)**
Operating Temperature	Under 70 C	94.5 C	**MISMATCH (CRITICAL)**
Power Supply Status	Stable	Low Voltage	**MISMATCH (CRITICAL)**

Detected Mismatches & Anomalies

1. Firmware Version Mismatch [CRITICAL]

Firmware version mismatch. Expected 'v9.4.0', found 'v9.0.2'.

2. PLC Version Mismatch [WARNING]

PLC configuration version mismatch. Expected 'v8.32', found 'v8.00'.

3. CPU Architecture Mismatch [CRITICAL]

Processor type mismatch. Reference specifies 'Intel Core i3-7100U' but found 'Intel Pentium 4415U'.

4. RAM (Memory) Mismatch [WARNING]

System memory allocation mismatch. Expected '16GB DDR4', active size is '8GB DDR4'.

5. Communication Ports Mismatch [CRITICAL]

Required port(s) offline: Profibus, COM2.

6. Installed Modules Mismatch [CRITICAL]

Critical expansion module(s) missing or failed: Power Logger Card.

7. Sensor Count Mismatch [WARNING]

Active sensor count mismatch. Expected '24' nodes, detected '20' nodes.

8. Operating Temperature Mismatch [CRITICAL]

CRITICAL: Operating temperature is dangerously high at 94.5 C. Immediate cooldown or shutdown required.

9. Power Supply Status Mismatch [CRITICAL]

CRITICAL: Power supply voltage has dropped below safety margin.

AI-Powered Diagnostics Analysis

Machine Safety & Health Assessment

DEGRADED / ACTION REQUIRED (Health: 0%): Mismatches or anomalous sensor readings detected. Operational risks present.

Root Cause Analysis

The diagnostic engine flagged the following configuration mismatches: - Firmware Version: Firmware version mismatch. Expected 'v9.4.0', found 'v9.0.2'. - PLC Version: PLC configuration version mismatch. Expected 'v8.32', found 'v8.00'. - CPU Architecture: Processor type mismatch. Reference specifies 'Intel Core i3-7100U' but found 'Intel Pentium 4415U'. - RAM (Memory): System memory allocation mismatch. Expected '16GB DDR4', active size is '8GB DDR4'. - Communication Ports: Required port(s) offline: Profibus, COM2. - Installed Modules: Critical expansion module(s) missing or failed: Power Logger Card. - Sensor Count: Active sensor count mismatch. Expected '24' nodes, detected '20' nodes. - Operating Temperature: CRITICAL: Operating temperature is dangerously high at 94.5 C. Immediate cooldown or shutdown required. - Power Supply Status: CRITICAL: Power supply voltage has dropped below safety margin. Typical root causes include: field component replacement with unauthorized modules, out-of-date PLC programming flashes, firmware regression during maintenance cycles, or thermal cooling degradation.

Suggested Maintenance Worksteps

1. Flush cooling heat-exchanger. Check coolant levels, auxiliary radiator fans, and verify hydraulic oil thermal status.
2. Connect Service Tool (CAT ET or Siemens TIA Portal) and flash target firmware and PLC image profiles matching reference configurations.
3. Audit active hardware modules. Replace generic CPU boards, RAM modules, or expansion cards with OEM-authorized parts.
4. Check supply transformer lines, inspect local power filter capacitor banks, and verify voltage regulator calibration.

Crucial Safety Notes

- Verify power source disconnect before opening PLC enclosures.
- WARNING: High temperature components! Allow machine to idle and cool down for 20 minutes before inspecting thermal zones.
- Ensure constant power backup is connected to the PLC logic controllers during the programming and flashing sequence.
- Follow Electrostatic Discharge (ESD) wrist strap procedures when handling exposed logic boards or memory chips.
- CRITICAL ELECTRICAL HAZARD: Wear insulated high-voltage gloves (Class 0) if measuring terminal rails directly.

Verification & Troubleshooting Steps

1. Calibrate thermistor sensors using an external infrared thermometer. Verify radiator fan relay controls.
2. Run firmware CRC checksum validation command. Check serial console output for boot loader warnings.

3. Perform full system diagnostic self-test from the service shell. Verify device tree registration of all local modules.
4. Use digital multimeter to measure main 24V DC bus voltage at full load and trace voltage drops.

PREPARED BY:	REVIEWED & SIGNED BY:
Automated Diagnostic Engine CAT Diagnostic Systems	Lead Engineer: John Administrator CAT Certified Field Inspector